

HARSH RAJ

Patna, Bihar, India

+91 7488638300 | harsh74raj88@gmail.com | LinkedIn:linkedin.com/in/harsh-raj-9981b428a

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science and Engineering (IoT) student with strong programming skills in Python and hands-on experience in embedded systems, IoT applications, automation projects, and sensor integration. Proficient in Python, C/C++, Arduino, ESP32. Experienced in developing real-world projects involving GPS tracking, automation, and hardware-software integration. Passionate about software development, problem-solving, and building scalable technology solutions.

EDUCATION

Bakhtiyarpur College of Engineering, Patna

Sept 2022 – May 2026

B.Tech in Computer Science and Engineering (Internet of Things)

CGPA: 6.9/10

Relevant Coursework: IoT, Embedded Systems, Cyber Security, Web Design Technologies

TECHNICAL SKILLS

Languages: Python, C, C++, HTML, CSS

Embedded & IoT: Arduino, ESP32, Microcontroller Programming, Sensor Interfacing, IoT Cloud

Protocols: I2C, Wi-Fi, Bluetooth, LoRa

Sensors/Modules: GPS Module, Vibration Sensor, OLED Display, Buzzer, Bluetooth Module

Tools: Arduino IDE, VS Code, Salesforce Trailhead

PROJECTS

IoT-Based Vehicle Accident Detection and Emergency Alert System

- Developed a vehicle-mounted accident detection system using GPS, accelerometer/vibration sensor and IoT communication.
- Designed crash-detection logic to identify sudden impacts and abnormal vehicle movement.
- Integrated GPS coordinates for location-based emergency alerts and real-time monitoring.
- Performed hardware interfacing, testing, debugging and embedded-system integration.

Bluetooth-Based Home Automation System

- Built an Arduino-based home automation system controlled through a mobile application.
- Implemented Bluetooth communication, relay switching and wireless automation logic.

INTERNSHIP EXPERIENCE

Polytropic Services Pvt. Ltd.

May 2025 – Jun 2025

Industrial Intern – Web Design Technologies

- Completed industrial training focused on HTML, CSS, JavaScript and web development workflow.

Salesforce Developer Virtual Internship

Dec 2023 – Jan 2024

- Completed Trailhead modules covering Apex, APIs, Process Automation, LWC, Testing and Debugging.

CERTIFICATIONS

Foundation of Cyber Security • Introduction to Ethical Hacking • Graphic Design with Photoshop • Event Management Certification

LEADERSHIP & ACTIVITIES

- Core Member and Event Management Lead, Google Developer Club.
- Coordinator, Hand of Hope Social Club and Flash Click Photography Club.
- Participated in Google Cloud Study Jam 2024, DevFest 2024, Solution Challenge Bootcamp 2025 and Google I/O Extended 2025.
- NSS Volunteer for social drives, blood donation camps and community initiatives.

LANGUAGES

English, Hindi